

Animate Asset Replacer

Recipe JSON Reference - v3.7.21

Reference for the key/value pairs used by the recipe JSON for embedded image and audio dataURI replacement. Relative replacement paths are resolved from the directory containing the recipe JSON.

1. Recipe Structure

A recipe contains global defaults plus a replacements array. Settings on an individual replacement override the corresponding global settings.

```
{
  "task": "replace_embedded_manifest_assets",
  "image_processing": { ... },
  "operation": { ... },
  "conditions": { ... },
  "verification": { ... },
  "replacements": [
    { ... },
    { ... }
  ],
  "output": {
    "type": "html",
    "filename_suffix": "_assets_replaced"
  }
}
```

2. Top-Level Keys

Key	Value / Type	Applies To	Meaning
task	string	Recipe	Use "replace_embedded_manifest_assets" for mixed image/audio recipes. Older image-only values remain supported.
image_processing	object	Images	Global image-processing defaults. Individual image entries can override these.
operation	object	Images / Audio	Global encoding/output defaults. Individual replacements can override these.
conditions	object	Images / Audio	Global conditions evaluated for each replacement.
verification	object	Images	Global verification options.
replacements	array	Images / Audio	One object per manifest asset to replace.
output	object	Recipe	Controls the generated HTML filename.

3. Replacement Entry Keys

Key	Value / Type	Applies To	Meaning
asset_id	string; required	Images / Audio	Manifest id to locate, e.g. "bg" or "spinSfx".

asset_type	"image" "audio"	Images / Audio	Explicitly identifies the replacement type. Recommended, especially for audio.
replacement_image	string path	Images	Replacement image file. Relative paths resolve from the recipe JSON directory.
replacement_audio	string path	Audio	Replacement audio file. Relative paths resolve from the recipe JSON directory.
image_processing	object	Images	Per-image overrides for global image_processing.
operation	object	Images / Audio	Per-asset overrides for global operation.
conditions	object	Images / Audio	Per-asset conditions, combined with global conditions.
verification	object	Images	Per-image verification overrides.

4. Image Processing Keys

Key	Value / Type	Applies To	Meaning
match_original_image_dimensions	boolean	Images	When true and resize_mode is not "fill", resize the replacement to the dimensions of the original embedded image.
resize_mode	"fill" "contain" "cover"	Images	Controls how Sharp fits the replacement when resizing. stretch maps to fill.
skip_resize_if_dimensions_already_match	boolean	Images	Avoids an unnecessary resize when source and original dimensions already match.
center	boolean	Images	Aligns the replacement image to the center of the original image assuming the original image registration point is {x: 0, y: 0}

5. Operation Keys

Key	Value / Type	Applies To	Meaning
preserve_replacement_format	boolean	Images	When true, keep the replacement image format when supported (PNG/JPEG/WebP).
mime_type	MIME string	Images / Audio	Explicit output MIME. Images commonly use "image/jpeg" or "image/png"; audio can use "audio/mpeg", "audio/wav", etc.
jpeg_quality	integer 1-100	Images	JPEG output quality. Defaults to 95 when not specified.
extract_original_image_dimensions	boolean	Images	Accepted recipe option; original dimensions are inspected automatically by the current processor.
data_uri_source	string	Images / Audio	Accepted recipe option; current processor obtains dataURI from the matched manifest entry.

convert_replacement_image_to_data_uri	boolean	Images	Accepted recipe option; conversion to Base64 dataURI is automatic.
replace_target_image_data	boolean	Images	Accepted recipe option; replacement is automatic once conditions pass.

6. Audio Behavior

Audio files are not transcoded in v0.3.2. The replacement file bytes are read as-is and embedded into the HTML as Base64. MIME type is selected from operation.mime_type when supplied; otherwise the app infers it from the file extension, falling back to the original embedded MIME type.

Key	Value / Type	Applies To	Meaning
.mp3 / .mpeg / .mpga	audio/mpeg	Audio	Automatically inferred MIME.
.wav	audio/wav	Audio	Automatically inferred MIME.
.ogg / .oga	audio/ogg	Audio	Automatically inferred MIME.
.m4a / .mp4	audio/mp4	Audio	Automatically inferred MIME.
.aac	audio/aac	Audio	Automatically inferred MIME.
.webm	audio/webm	Audio	Automatically inferred MIME.

7. Conditions

Key	Value / Type	Applies To	Meaning
asset_id_equals	string	Images / Audio	Only replace when the current manifest asset id equals this value.
asset_type_equals	"image" "audio"	Images / Audio	Only replace when the detected/declared asset type matches.
replacement_width_greater_than	number	Images	Replacement image width must be greater than this value.
replacement_width_at_least	number	Images	Replacement image width must be at least this value.
replacement_height_at_least	number	Images	Replacement image height must be at least this value.
replacement_format_in	array of strings	Images	Allowed Sharp image formats, e.g. ["png", "jpeg"].
original_width_equals	number	Images	Original embedded image width must equal this value.
original_height_equals	number	Images	Original embedded image height must equal this value.
replacement_audio_bytes_at_least	number	Audio	Replacement audio file size must be at least this many bytes.
replacement_audio_extensions_in	array of strings	Audio	Allowed extensions, including the leading dot, e.g. [".mp3", ".wav"].

8. Verification Keys

Key	Value / Type	Applies To	Meaning
verify_encoded_image_dimensions_match_original	boolean	Images	After encoding, verifies that output image width/height match the original embedded image.

9. Output Keys

Key	Value / Type	Applies To	Meaning
type	"html"	Recipe	Output document type.
filename_suffix	string	Recipe	Appended to the source HTML filename, e.g. "_assets_replaced".

10. Image Example

```
{
  "task": "replace_embedded_manifest_assets",
  "image_processing": {
    "match_original_image_dimensions": true,
    "resize_mode": "stretch",
    "skip_resize_if_dimensions_already_match": true
  },
  "operation": {
    "preserve_replacement_format": true
  },
  "replacements": [
    {
      "asset_id": "bg",
      "asset_type": "image",
      "replacement_image": "new_bg.jpg"
    },
    {
      "asset_id": "logo",
      "asset_type": "image",
      "replacement_image": "images/new_logo.png",
      "operation": {
        "preserve_replacement_format": true
      },
      "conditions": {
        "replacement_width_at_least": 200
      }
    }
  ],
  "output": {
    "type": "html",
    "filename_suffix": "_images_replaced"
  }
}
```

11. Audio Example

```
{
  "task": "replace_embedded_manifest_assets",
  "replacements": [
    {
      "asset_id": "spinSfx",
      "asset_type": "audio",
      "replacement_audio": "audio/spin.mp3",
      "operation": {
        "mime_type": "audio/mpeg"
      }
    },
    {
      "asset_id": "winSfx",
      "asset_type": "audio",
      "replacement_audio": "audio/win.wav",
      "conditions": {
        "replacement_audio_extension_in": [".wav"]
      }
    }
  ],
}
```

```

"output": {
  "type": "html",
  "filename_suffix": "_audio_replaced"
}
}

```

12. Mixed Image + Audio Example

```

{
  "task": "replace_embedded_manifest_assets",
  "image_processing": {
    "match_original_image_dimensions": true,
    "resize_mode": "stretch"
  },
  "replacements": [
    {
      "asset_id": "bg",
      "asset_type": "image",
      "replacement_image": "new_bg.jpg"
    },
    {
      "asset_id": "spinSfx",
      "asset_type": "audio",
      "replacement_audio": "spin.mp3",
      "operation": {
        "mime_type": "audio/mpeg"
      }
    }
  ],
  "output": {
    "type": "html",
    "filename_suffix": "_assets_replaced"
  }
}

```

13. Custom JavaScript Conditions

The file `src/rules/conditions.js` runs once per replacement after JSON conditions. Return true to perform the replacement, false to skip it, or throw an Error to stop processing.

```

async function shouldReplace(context) {
  if (
    context.asset.type === "audio" &&
    context.replacementAudio.bytes > 5 * 1024 * 1024
  ) {
    throw new Error("Replacement audio must be 5 MB or smaller.");
  }

  return true;
}

```

14. Path Resolution

A relative path such as `"new_bg.jpg"`, `"audio/spin.mp3"`, or `"images/logo.png"` is resolved relative to the folder containing the selected recipe JSON. Absolute paths are used unchanged.

15. Notes on Current v0.3.2 Implementation

The schema accepts a few legacy/configuration keys that are not independently acted upon by the current processor. These are identified in the tables above. The keys that directly control current behavior are `asset_id`, `asset_type`, `replacement_image/replacement_audio`, `image_processing.match_original_image_dimensions`, `resize_mode`,

skip_resize_if_dimensions_already_match, operation.preserve_replacement_format, operation.mime_type, operation.jpeg_quality, the implemented conditions, verification, and output.filename_suffix.